

## Abstract

Through this observing program we will be obtaining the rotational period of a given asteroid and comparing our measured period with archival data found online. We will be attempting to follow observing plans similar to those set before us, so it will be interesting to see what results we get using those similar methods. What also makes this program interesting is that rotational periods cannot be directly measured; they must be calculated using light curves. Our targets will be the main-belt asteroids, particularly those that have rotational periods under 6 hours and visible magnitudes of 12-14  $V_{mag}$ . The targets must also be slow enough for non-sidereal tracking, as the telescope has a slew-rate limit. As of now, our main target is 155 Scylla.

Our observations will be 60 second photometric exposures taken with the Mont4k CCD mounted to the Kuiper 61" telescope using the I-band filter. In these exposures, light from the asteroid will be contained to some set of pixels. To generate an accurate light curve, the light from an asteroid must be calibrated with some reference background star. However, instead of having their light be contained to some pixel set, the star's light will instead "streak" across the exposure. To get the total flux from that star, the pixels within the streak must have their values added. The asteroid then must also have a slow enough angular speed across the sky such that the entire streak of the reference star is captured within the image.

By getting rotational periods using light curves, we may also uncover some additional information about the asteroid. Density, shape, and size are all traits that could be obtained from our observations. These will be interesting to see how the analysis of our findings matches real-world data

## Tolerable Moon Brightness

Our observations should be able to tolerate a fairly bright moon, as long as our target asteroid is at a sizeable angular separation from the moon. The brighter the moon, the larger the angle must be. If the moon is in fact too bright, it could oversaturate our images. If the saturation from the moon is also imbalanced across all the pixels, then our relative brightness measurements will be off. We will be referencing each asteroid measurement with the same background star also found in the image. If there's an uneven saturation due to the moon, then the pixels containing the reference star or asteroid could be more or less saturated in comparison to one another, giving us an inaccurate light curve. Thus, a brighter moon could give us imprecise measurements.

**Telescope Requested - Kuiper 61"**

**Instrument - Mont4K**

## Potential Target List

| Object Name | Right Ascension | Declination | Apparent V magnitude | Object Type | Visibility |
|-------------|-----------------|-------------|----------------------|-------------|------------|
| 155 Scylla  | 00 51 54.10     | -04 19 27.3 | 14.167               | Asteroid    | Bi-yearly  |
| 170 Maria   | 22 50 43.81     | +13 26 04.4 | 13.209               | Asteroid    | Bi-yearly  |

|                |             |             |        |          |           |
|----------------|-------------|-------------|--------|----------|-----------|
| 188 Menippe    | 03 02 24.26 | +24 23 46.2 | 13.391 | Asteroid | Bi-yearly |
| 167 Urda       | 01 36 58.47 | +07 56 15.8 | 13.302 | Asteroid | Bi-yearly |
| 248 Lameia     | 02 33 32.76 | +18 50 51.8 | 14.063 | Asteroid | Bi-yearly |
| 295 Theresia   | 01 39 24.80 | +14 59 13.3 | 13.162 | Asteroid | Bi-yearly |
| 2 Pallas       | 01 19 37.37 | -14 59 29.9 | 8.265  | Asteroid | Bi-yearly |
| 236 Honoria    | 23 05 13.51 | -04 08 34.1 | 11.4   | Asteroid | Bi-yearly |
| 71 Niobe       | 23 12 31.07 | +22 53 29.8 | 11.9   | Asteroid | Bi-yearly |
| 695 Bella      | 23 31 20.56 | +24 18 41.0 | 12.0   | Asteroid | Bi-yearly |
| 106 Dione      | 23 41 02.37 | -08 42 08.8 | 11.8   | Asteroid | Bi-yearly |
| 918 Itha       | 01 16 01.18 | +25 32 34.5 | 14.0   | Asteroid | Bi-yearly |
| 321 Florentina | 14 35 10.12 | -15 31 53.2 | 15.982 | Asteroid | Bi-yearly |

**Note:**

There are over 50 asteroids that meet the visibility constraints for the night that we have been assigned. If for some reason the night changes, then we will either have to pick new targets, or these targets will be in different locations.

## Experimental Design & Technical Description

### 1. SNR and Time Requirements

- a. We need an SNR of at least 50 to get an accurate light curve of the asteroid. However, we're aiming for an SNR of at least 150. Based on Haitao Huang *et al* 2026, we will be using an exposure time of 60 seconds in the I-band. Each

asteroid will need to be observed for double its period, so for 155 Scylla – which has a period of ~7.95 hours – we would need to observe for roughly 16 hours.

## **2. Connection with Abstract**

- a. The observations will produce scatterplots of time versus flux. During the asteroid's rotation period the relative flux measured should change in a repeated fashion, producing a sinusoidal curve. We take the average time for the asteroid to return to its initial measured flux over supposedly a rotational period and a half and take that as our final measurement period value. With each image, we are comparing the flux to the same reference stars to find the relative brightness throughout the rotation. By measuring the changes in this light curve over time, we are able to uncover the science of rotational periods and asteroid elongation/shape.

## **3. Back-up Plan**

- a. If we cannot observe, we will attain light curve or flux measurements from online archival data repositories from the Planetary Data System website. We would take data from one of our potential targets and process the data as we otherwise would from there.
  - i. <https://sbn.psi.edu/pds/resource/lc.html>
  - ii. <https://iopscience.iop.org/article/10.1088/1538-3873/ae7cb6/pdf>